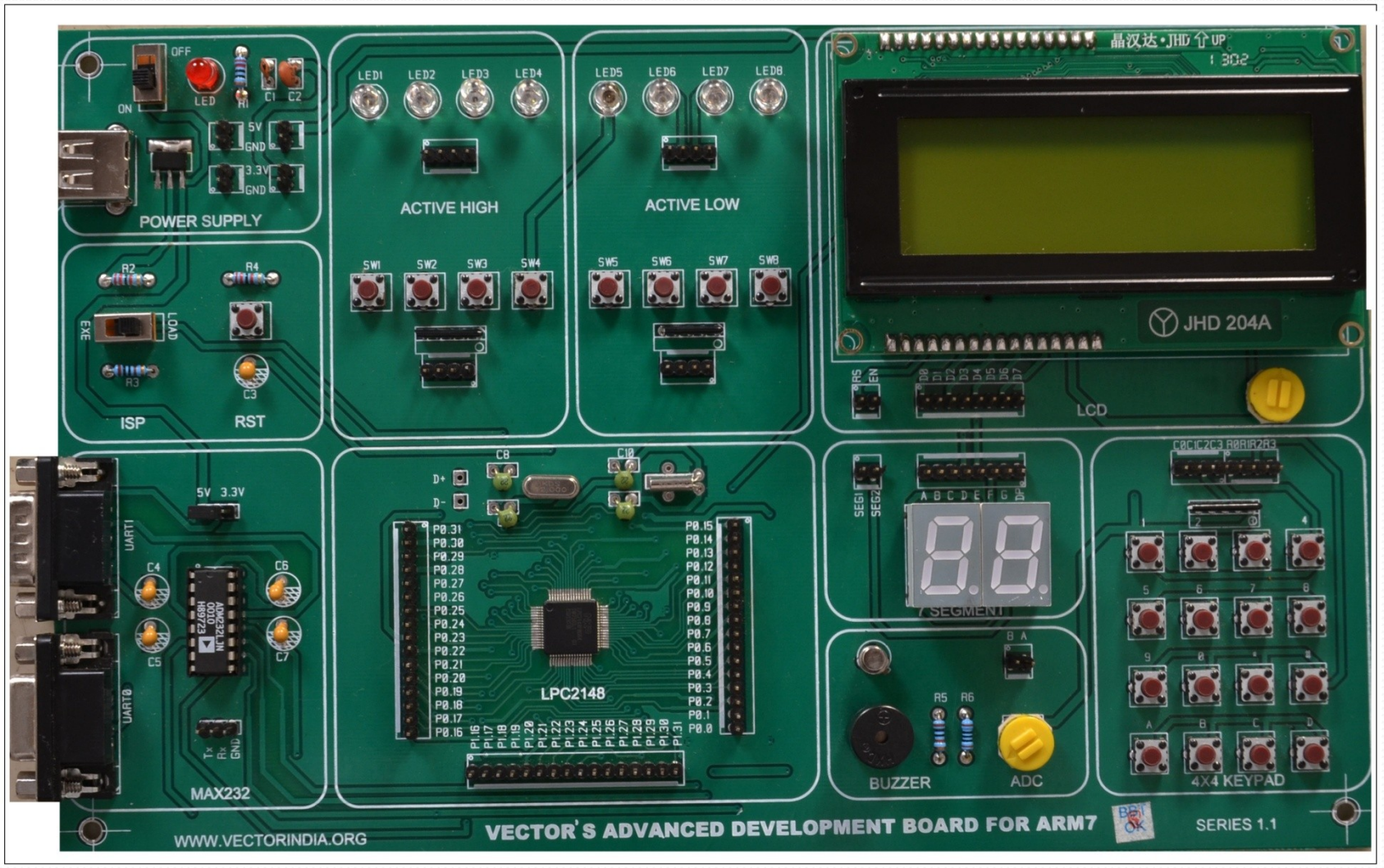
VECTOR'S LPC2148 DEVELOPMENT BOARD

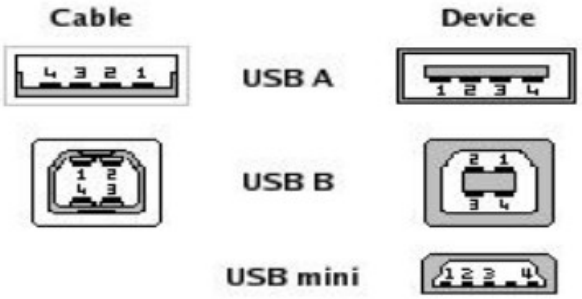
LPC 2148


POWER SUPPLY
ACTIVE HIGH
ACTIVE LOW
ISP
RST
LCD
JHD 204A
7 SEGMENT
4X4 KEYPAD
BUZZER
ADC
MAX232
LPC2148
WWW.VECTORINDIA.ORG
VECTOR'S ADVANCED DEVELOPMENT BOARD FOR ARM7
SERIES 1.1

POWER SUPPLY

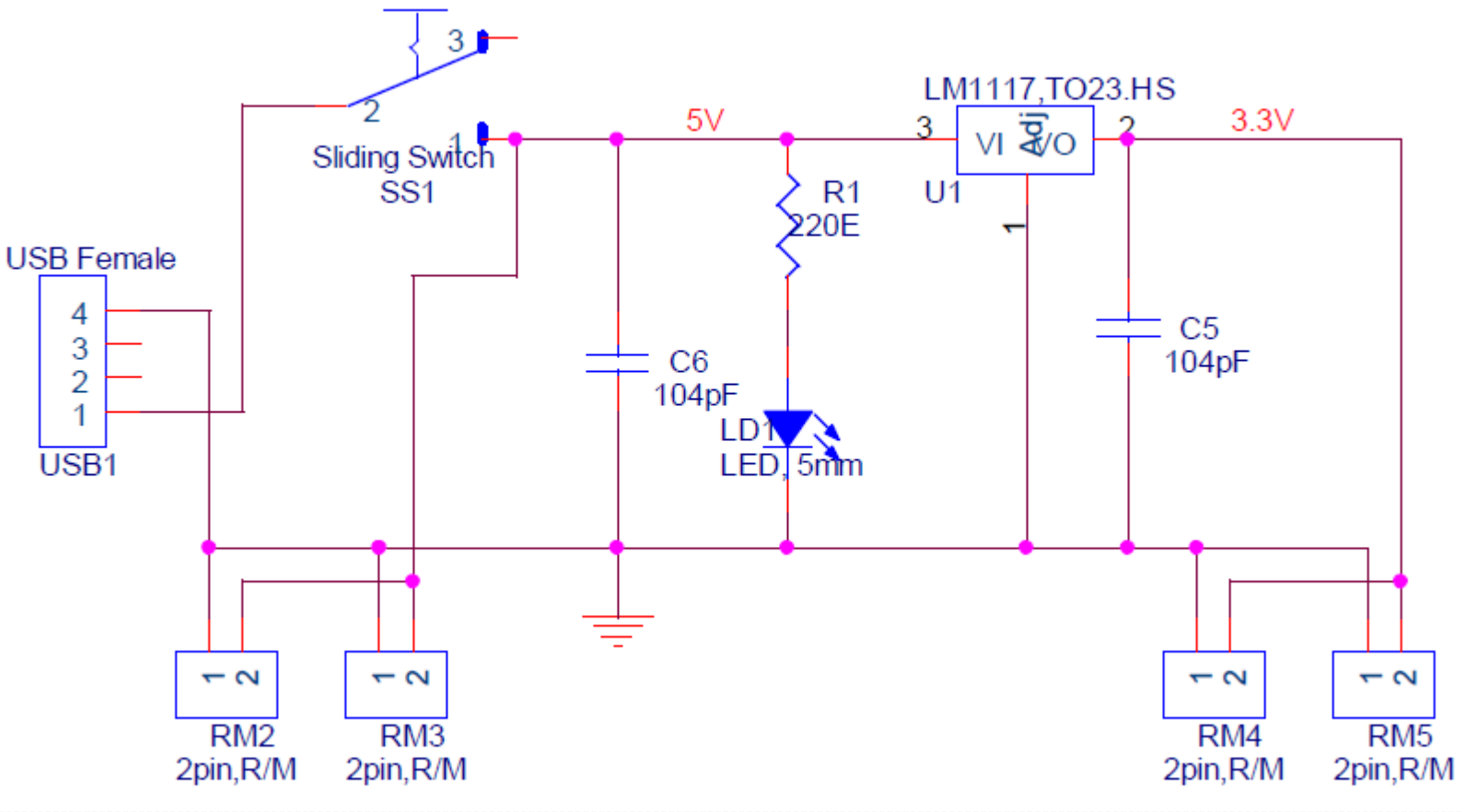


USB INTERFACING



Pin	Signal	Color	Description
1	VCC		+5V
2	D-		Data -
3	D+		Data +
4	GND		Ground

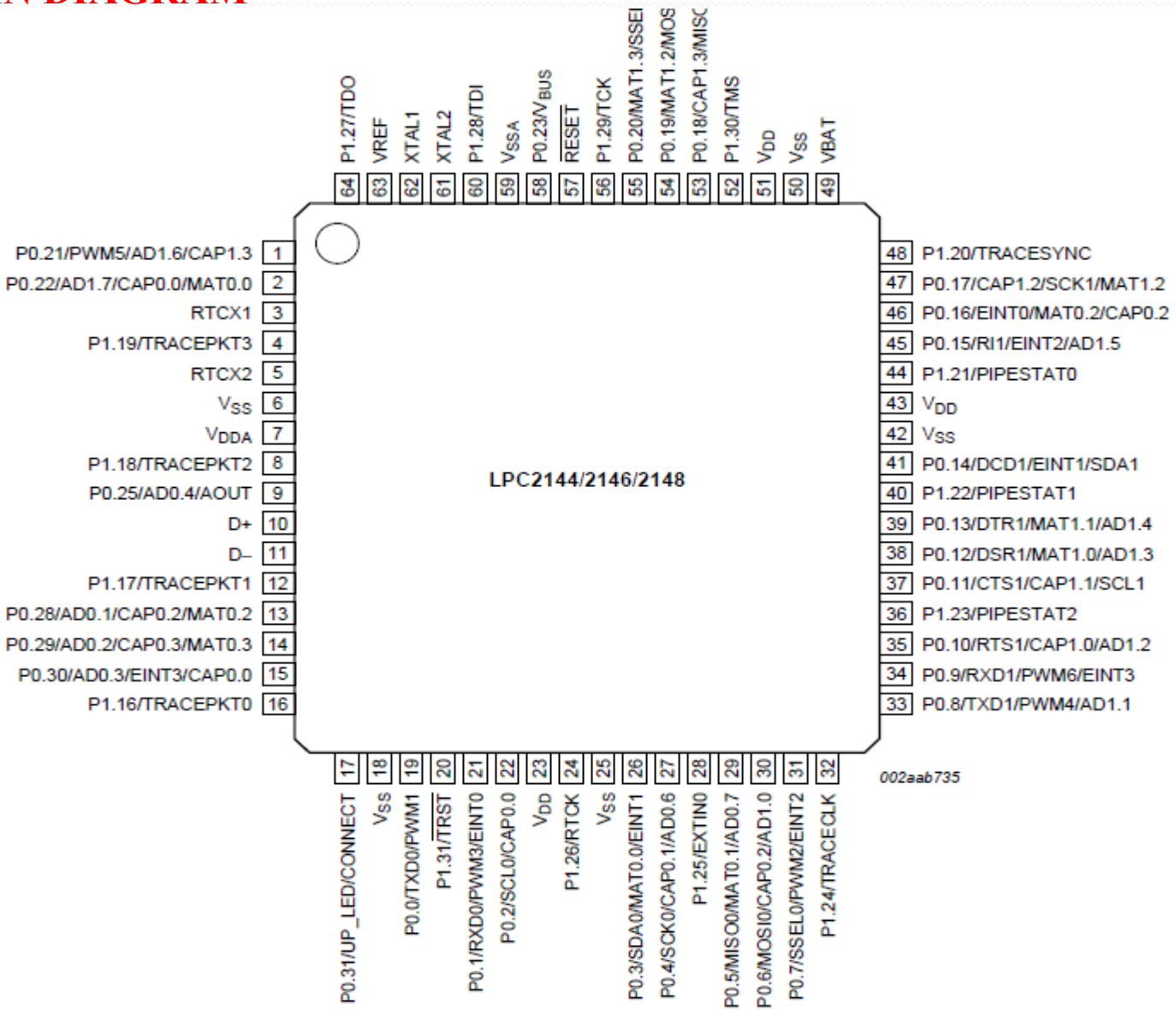
POWER SUPPLY



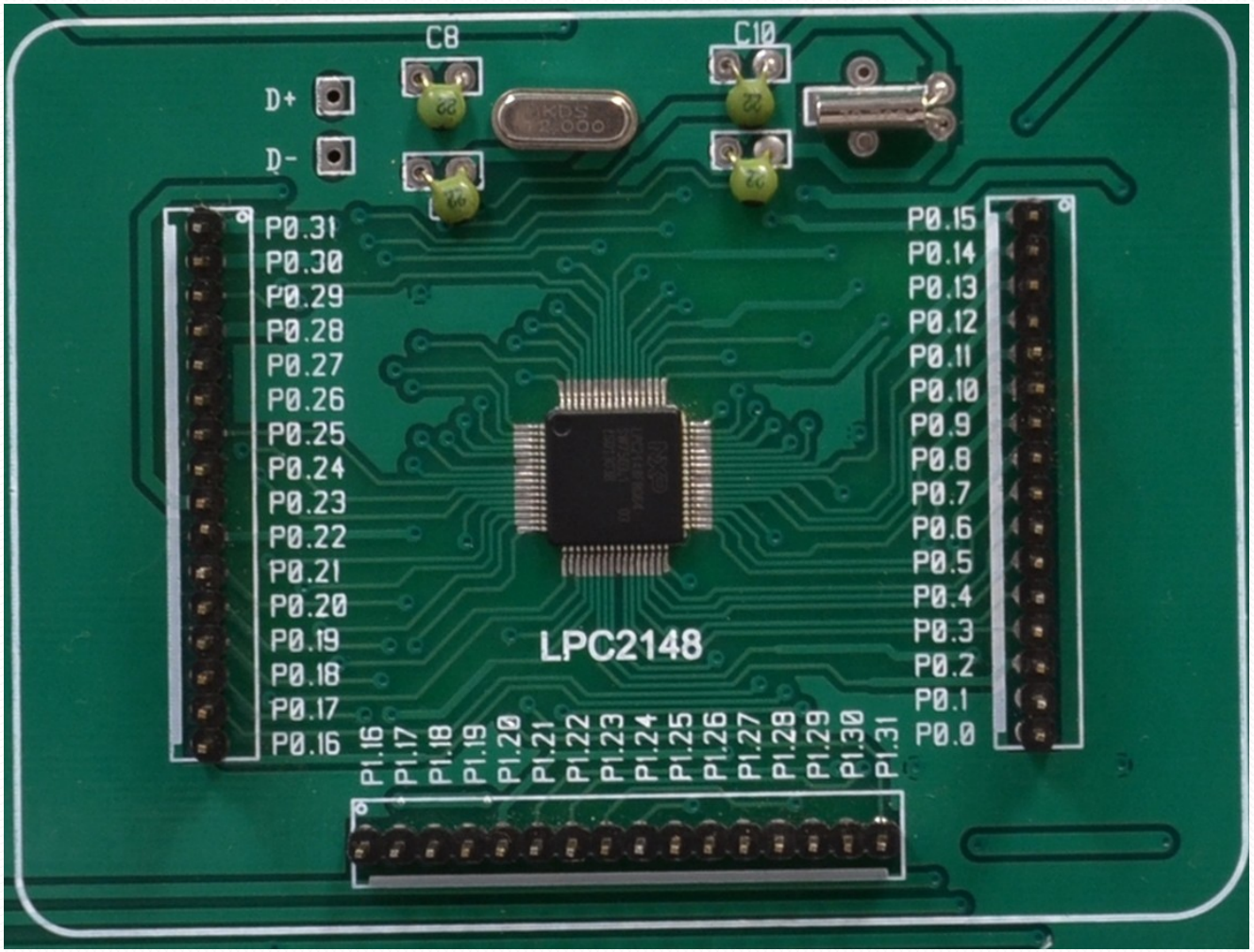
FEATURES

- 16-bit/32-bit ARM7 microcontroller in a tiny LQFP64 package
- CPU operating voltage range of 3.0 V to 3.6 V
- 5 V DC ---FOR LCD AND MAX232
- 45 GPIO(P0.0----P0.31 & P1.16----P1.31)
- P0.31 O/P PIN ONLY, P0.24, P0.26 and P0.27 are not available.
- On-chip integrated oscillator operates with an external crystal from 1 MHz to 25 MHz.
- Two UART, Two I2C,One SPI
- Low power Real-Time Clock (RTC) with independent power and 32.768 KHz clock input.

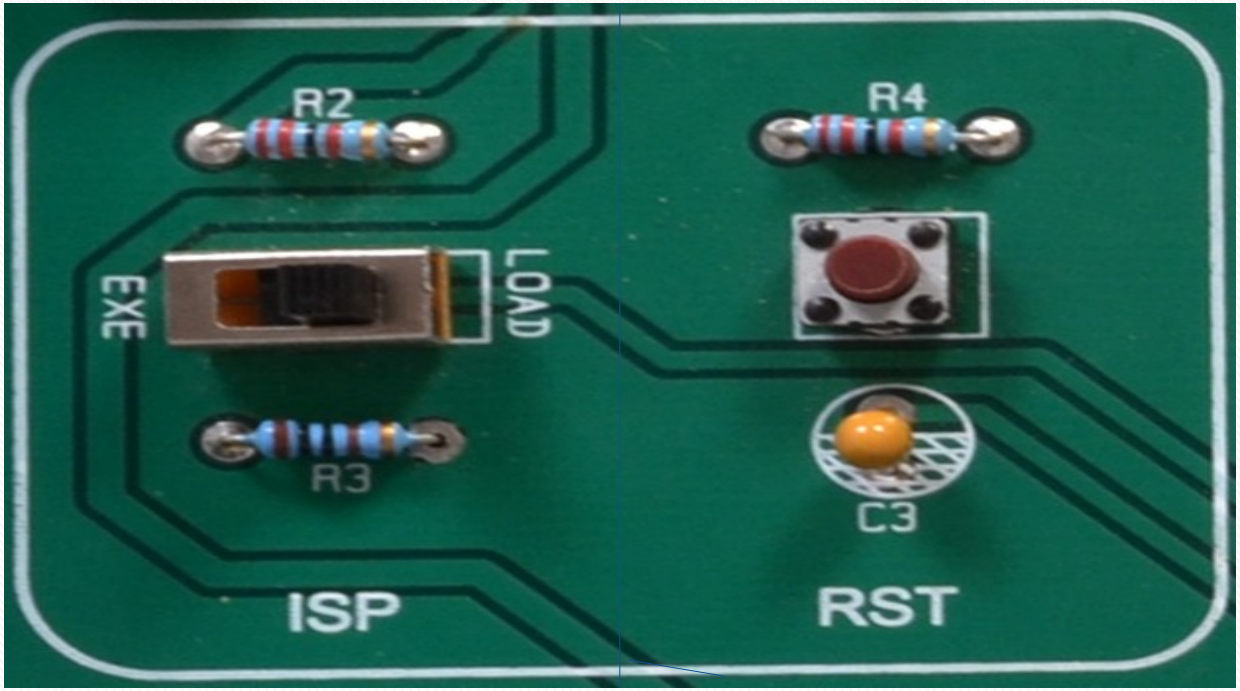
PIN DIAGRAM


LPC2144/2146/2148
1 P0.21/PWM5/AD1.6/CAP1.3
2 P0.22/AD1.7/CAP0.0/MAT0.0
3 RTCX1
4 P1.19/TRACEPKT3
5 RTCX2
6 VSS
7 VDDA
8 P1.18/TRACEPKT2
9 P0.25/AD0.4/AOUT
10 D+
11 D−
12 P1.17/TRACEPKT1
13 P0.28/AD0.1/CAP0.2/MAT0.2
14 P0.29/AD0.2/CAP0.3/MAT0.3
15 P0.30/AD0.3/EINT3/CAP0.0
16 P1.16/TRACEPKT0
17 P0.31/UP_LED/CONNECT
18 VSS
19 P0.0/TXD0/PWM1
20 P1.31/TRST
21 P0.1/RXD0/PWM3/EINT0
22 P0.2/SCL0/CAP0.0
23 VDD
24 P1.26/RTCK
25 VSS
26 P0.3/SDA0/MAT0.0/EINT1
27 P0.4/SCK0/CAP0.1/AD0.6
28 P1.25/EXTIN0
29 P0.5/MISO0/MAT0.1/AD0.7
30 P0.6/MOSI0/CAP0.2/AD1.0
31 P0.7/SSEL0/PWM2/EINT2
32 P1.24/TRACECLK
33 P0.8/TXD1/PWM4/AD1.1
34 P0.9/RXD1/PWM6/EINT3
35 P0.10/RTS1/CAP1.0/AD1.2
36 P1.23/PIPESTAT2
37 P0.11/CTS1/CAP1.1/SCL1
38 P0.12/DSR1/MAT1.0/AD1.3
39 P0.13/DTR1/MAT1.1/AD1.4
40 P1.22/PIPESTAT1
41 P0.14/DCD1/EINT1/SDA1
42 VSS
43 VDD
44 P1.21/PIPESTAT0
45 P0.15/RI1/EINT2/AD1.5
46 P0.16/EINT0/MAT0.2/CAP0.2
47 P0.17/CAP1.2/SCK1/MAT1.2
48 P1.20/TRACESYNC
49 VBAT
50 VSS
51 VDD
52 P1.30/TMS
53 P0.18/CAP1.3/MISO
54 P0.19/MAT1.2/MOSI
55 P0.20/MAT1.3/SSEL
56 P1.29/TCK
57 RESET
58 P0.23/VBUS
59 VSSA
60 P1.28/TDI
61 XTAL2
62 XTAL1
63 VREF
64 P1.27/TDO
002aab735

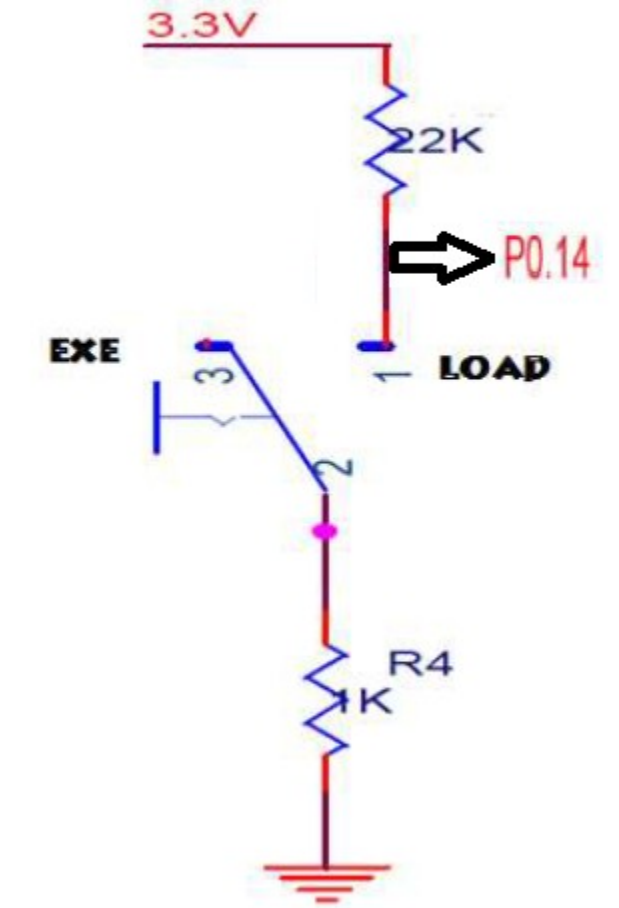
LPC 2148


C8
C10
D+
D-
P0.31
P0.30
P0.29
P0.28
P0.27
P0.26
P0.25
P0.24
P0.23
P0.22
P0.21
P0.20
P0.19
P0.18
P0.17
P0.16
P0.15
P0.14
P0.13
P0.12
P0.11
P0.10
P0.9
P0.8
P0.7
P0.6
P0.5
P0.4
P0.3
P0.2
P0.1
P0.0
LPC2148
P1.16
P1.17
P1.18
P1.19
P1.20
P1.21
P1.22
P1.23
P1.24
P1.25
P1.26
P1.27
P1.28
P1.29
P1.30
P1.31

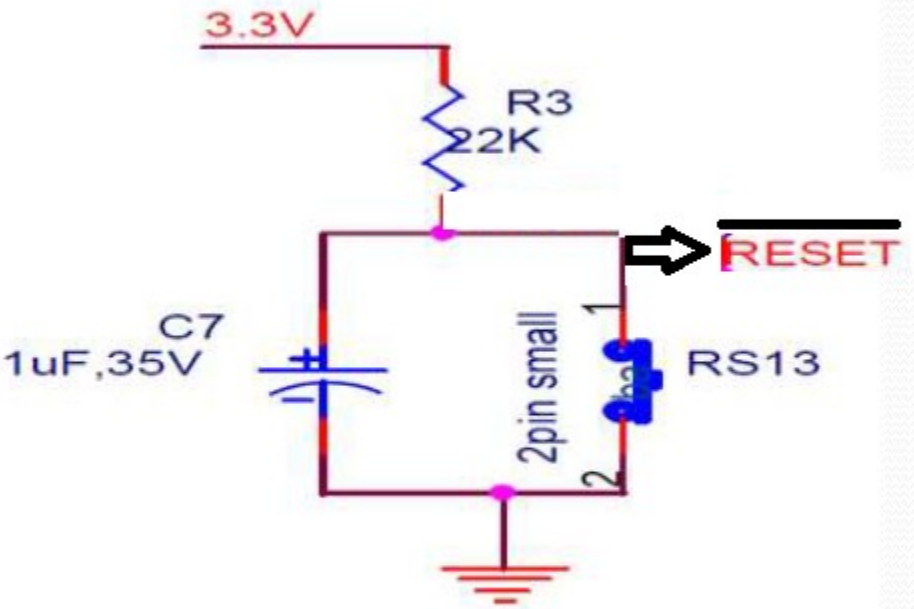
ISP **RESET**



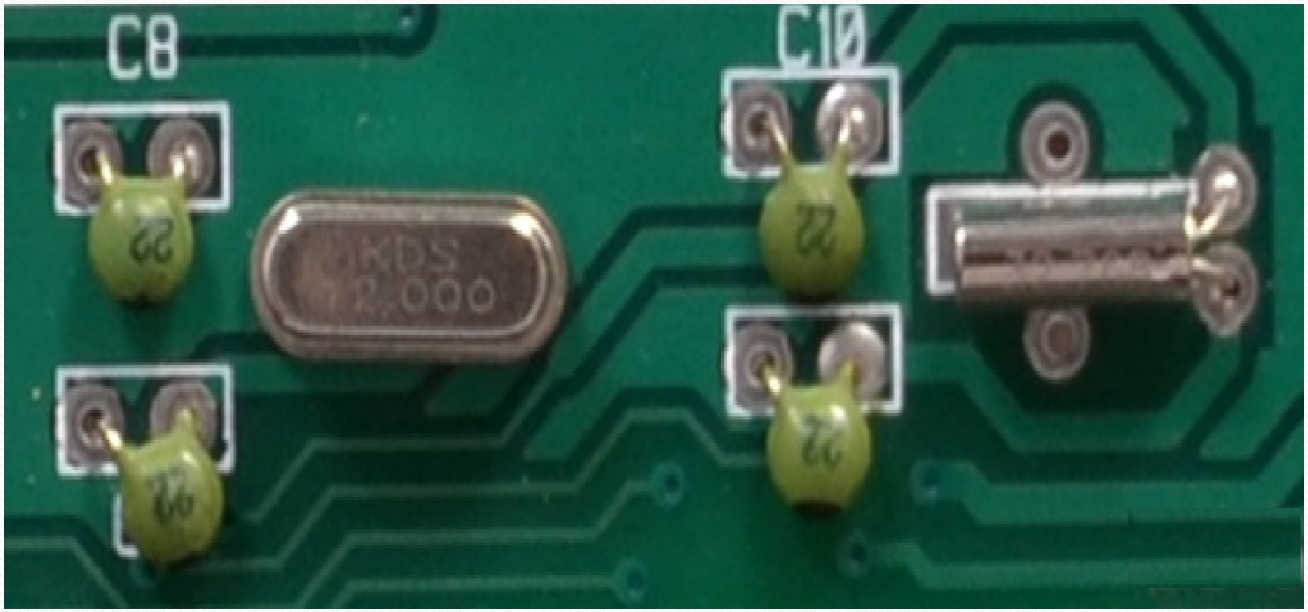
ISP


3.3V
22K
P0.14
EXE
3
1
LOAD
2
R4
1K

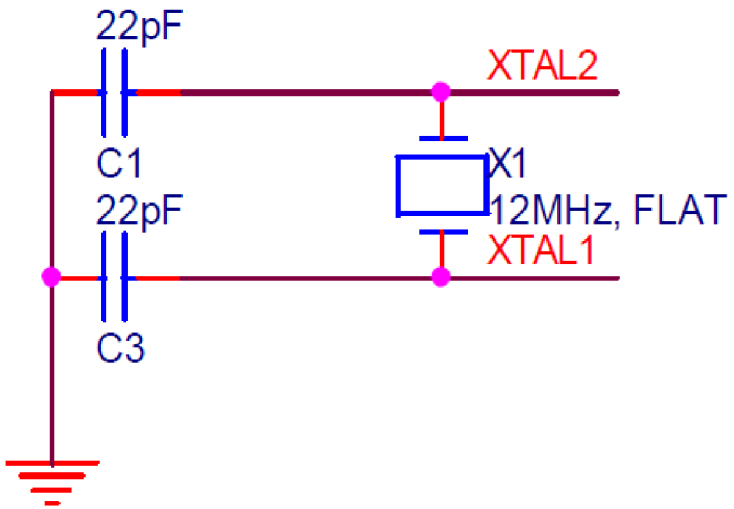
RESET

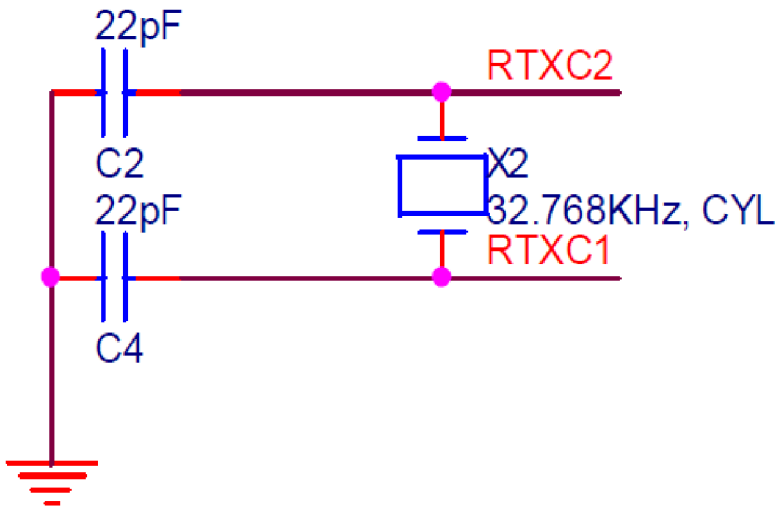

3.3V
R3
22K
RESET
C7
1uF,35V
2pin small
1
2
RS13

CRYSTAL'S

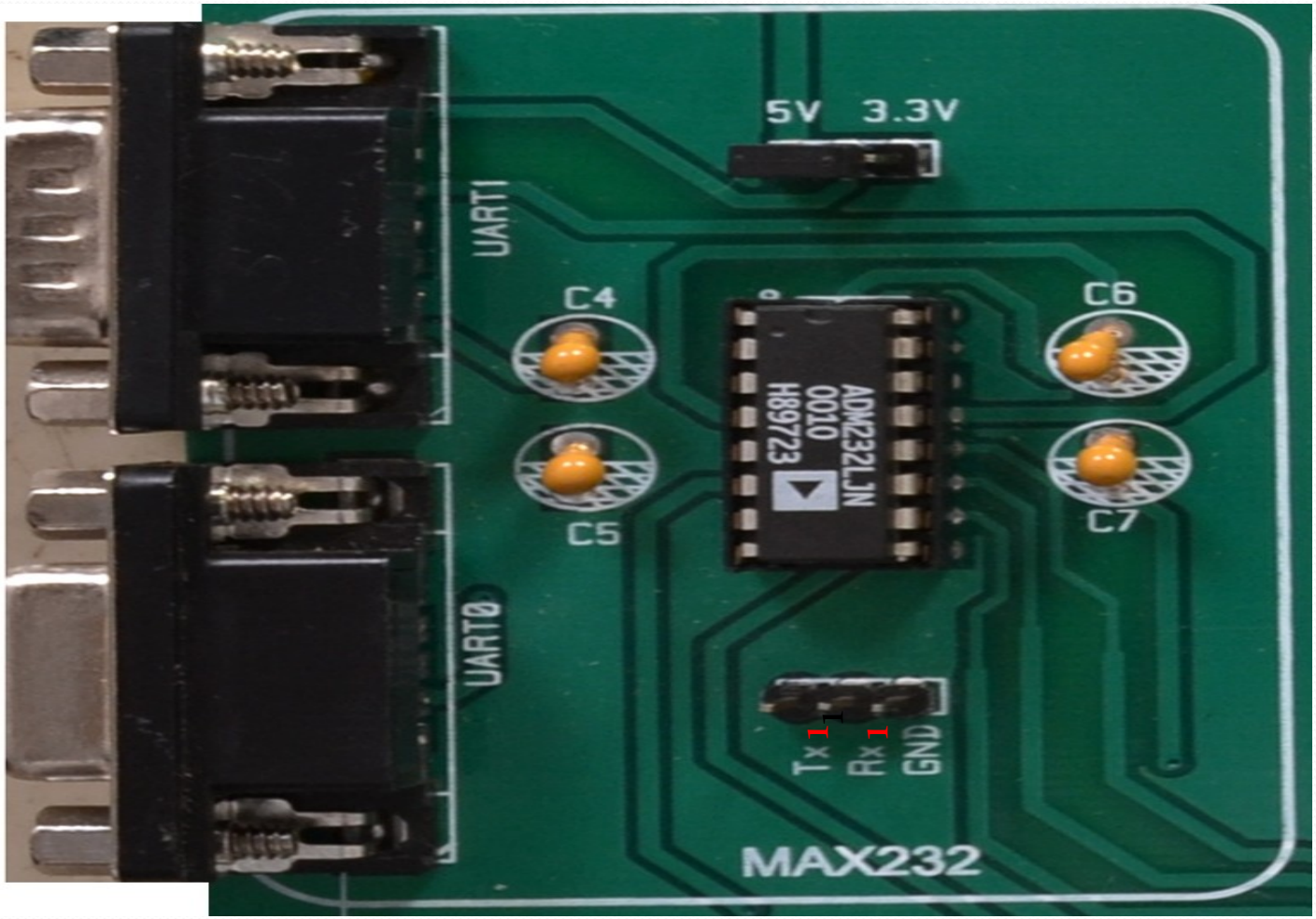


CRYSTAL'S

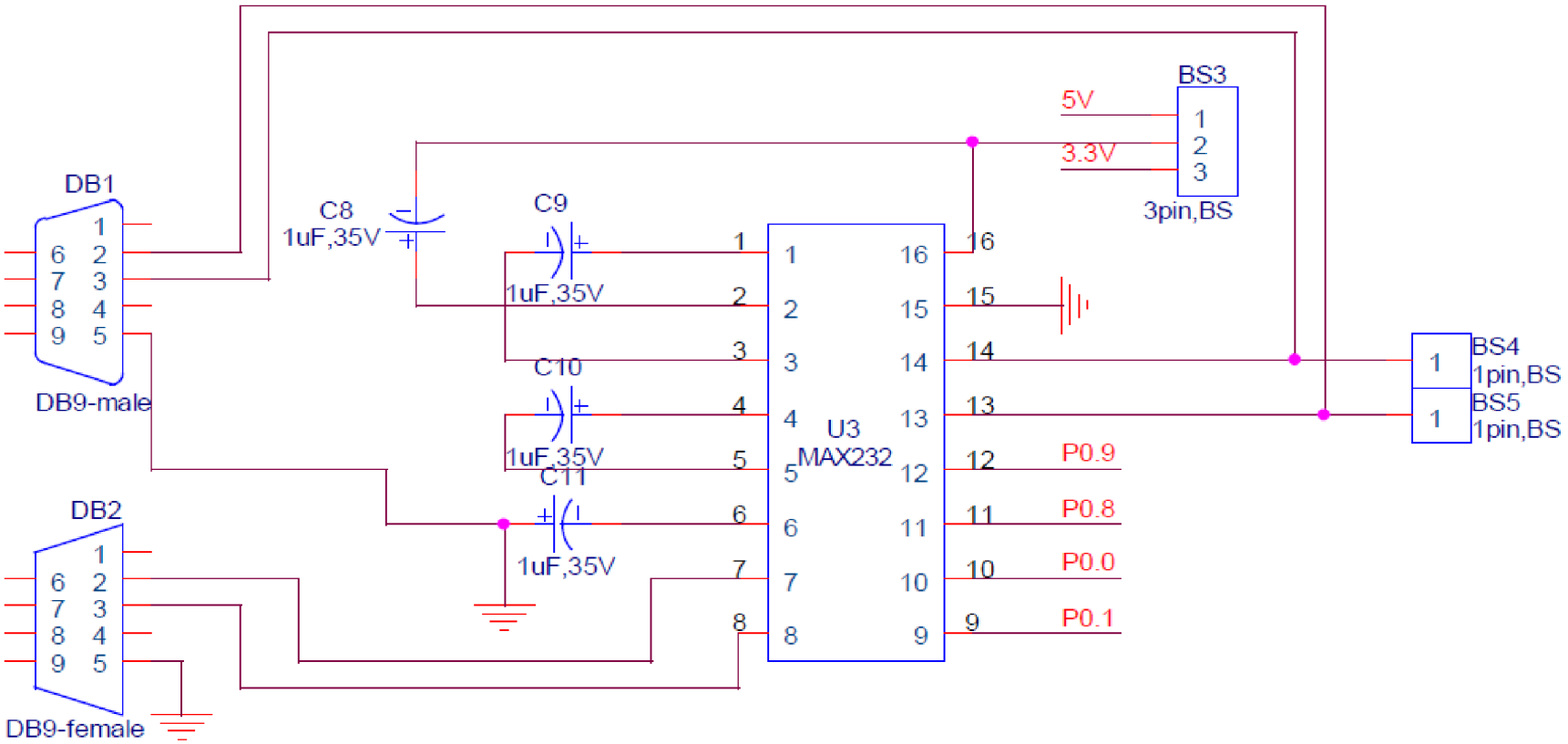

22pF
C1
22pF
C3
XTAL2
X1
12MHz, FLAT
XTAL1


22pF
C2
22pF
C4
RTXC2
X2
32.768KHz, CYL
RTXC1

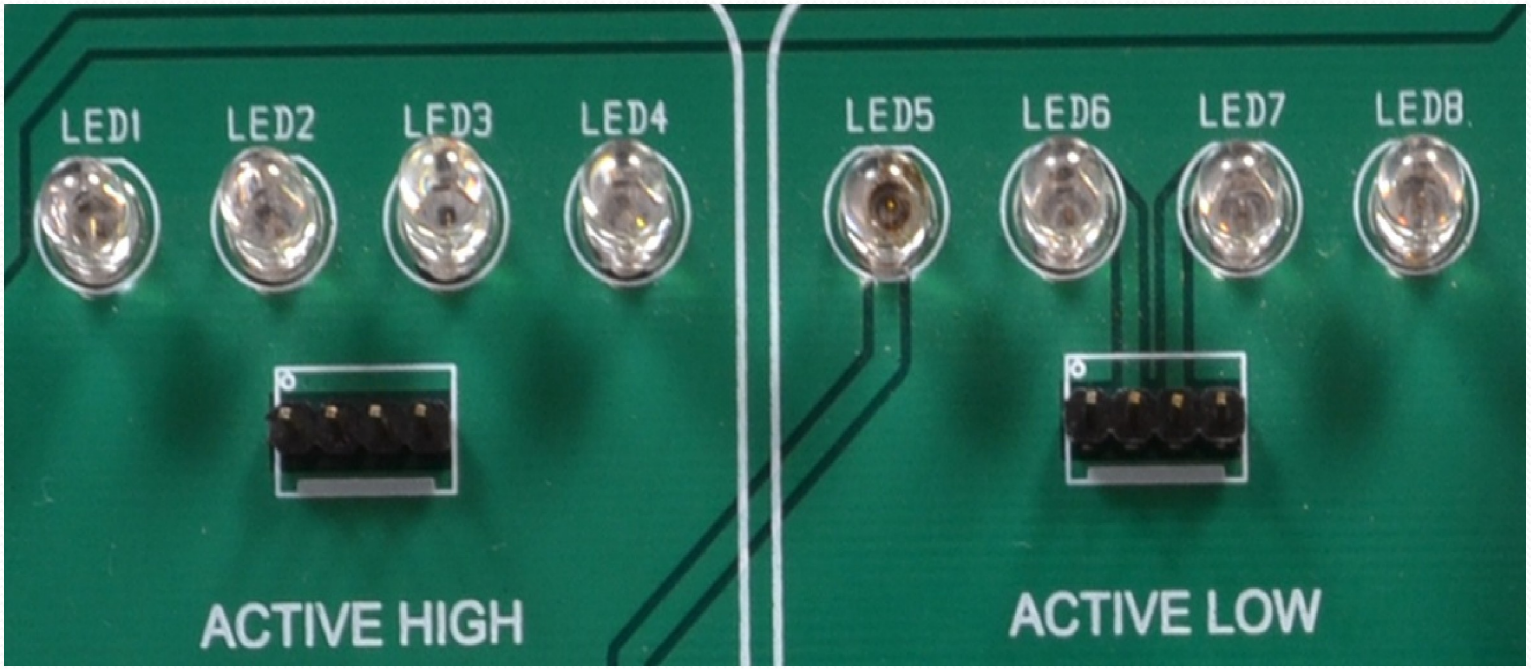
MAX 232


5V 3.3V
UART1
C4
C6
C5
C7
ADM232LJN
0010
H8972
UART0
Tx 1
Rx 1
GND
MAX232

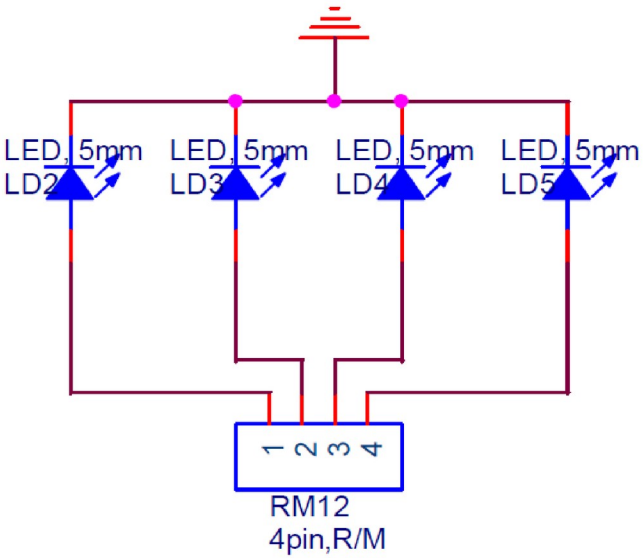
MAX 232


DB1
DB9-male
DB2
DB9-female
C8
1uF,35V
C9
1uF,35V
C10
1uF,35V
C11
1uF,35V
U3
MAX232
BS3
5V
3.3V
3pin,BS
BS4
1pin,BS
BS5
1pin,BS
P0.9
P0.8
P0.0
P0.1

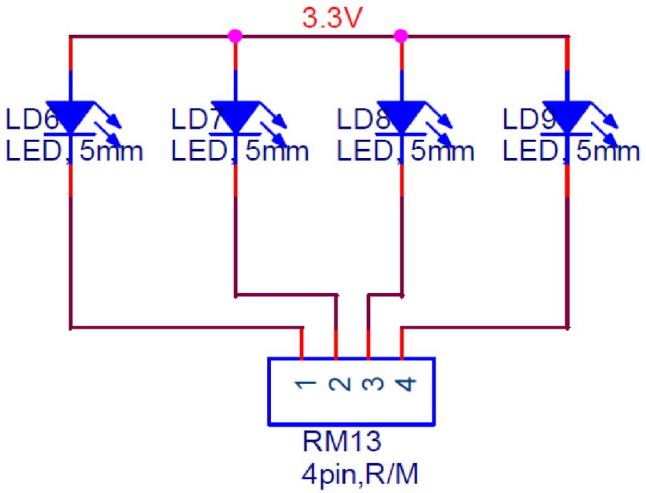
LED’S


LED1
LED2
LED3
LED4
LED5
LED6
LED7
LED8
ACTIVE HIGH
ACTIVE LOW

LED’S

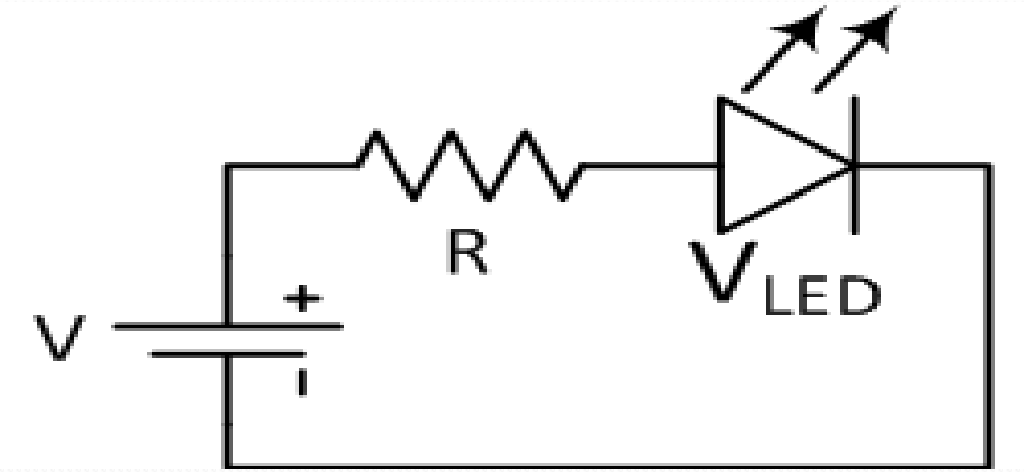


ACTIVE HIGH



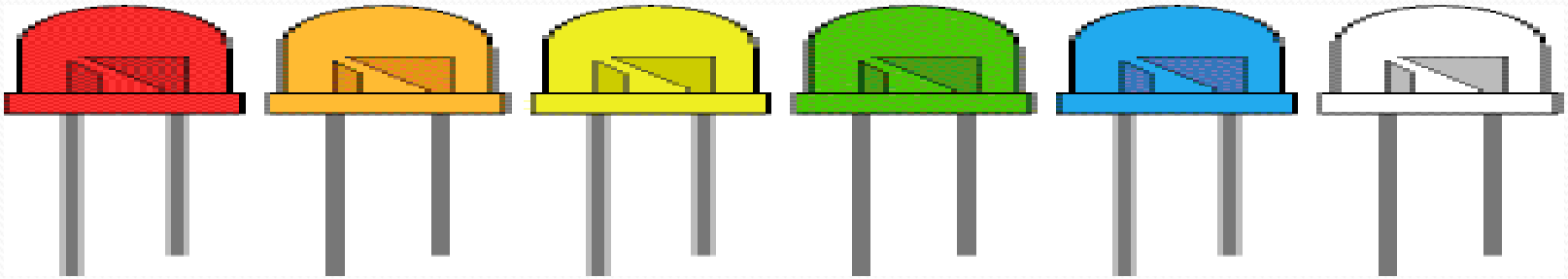
ACTIVE LOW

HOW TO FIND RESISTOR VALUE



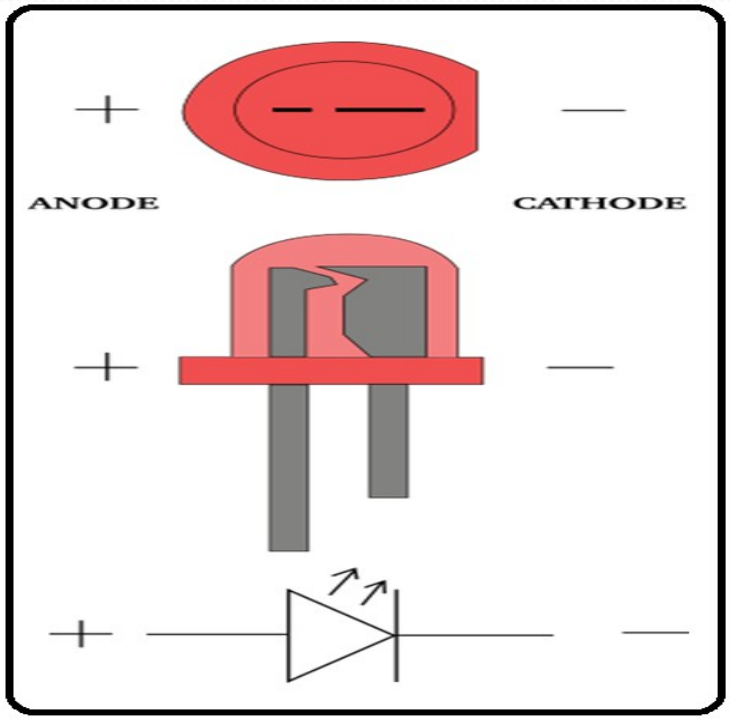
RESISTOR = (INPUT VOLTAGE – LED VOLTAGE) / LED CURRENT

LED'S :

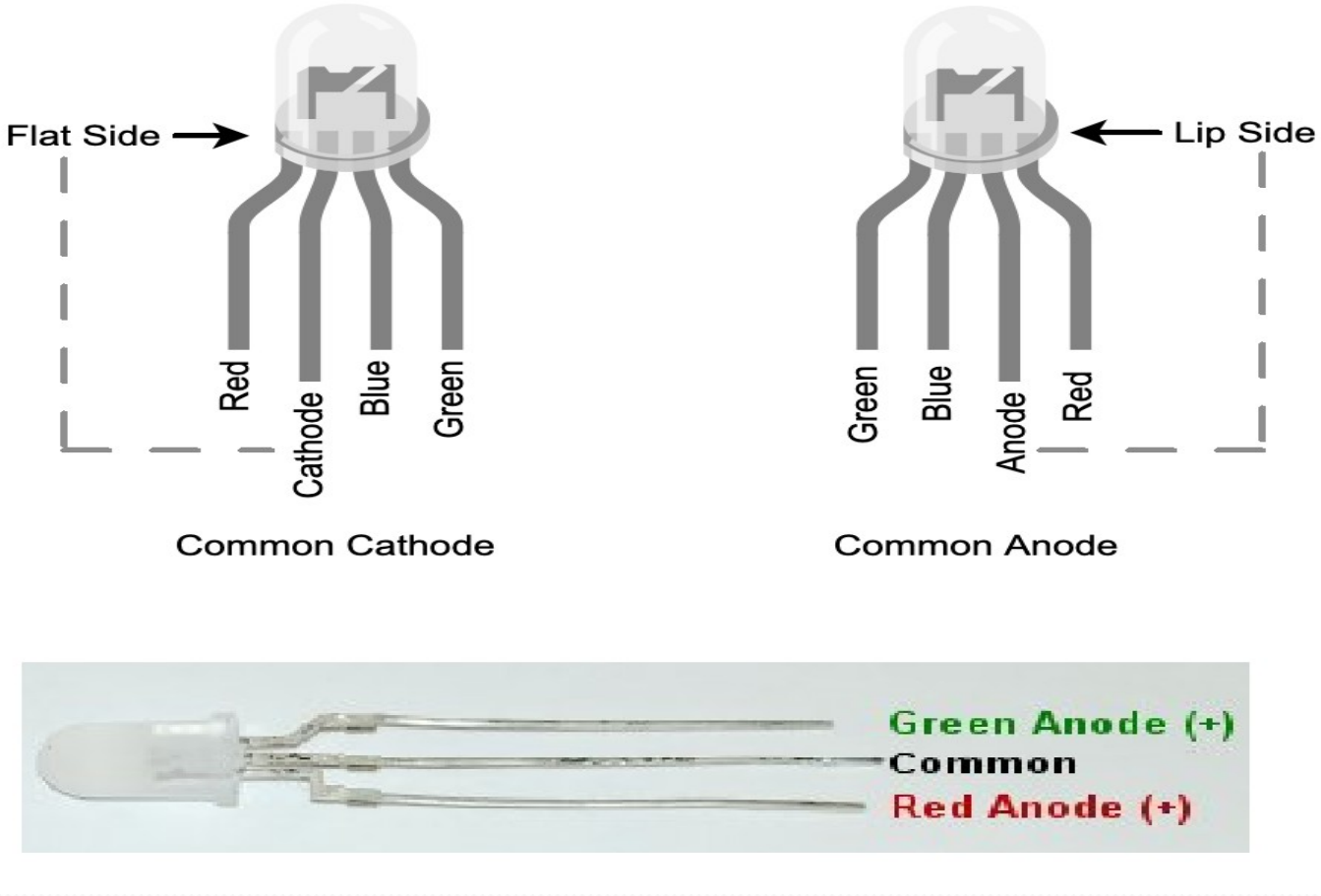


LED Colour	Forward Voltage V_f	Forward Current I_f
White	3.2 V to 3.8 V	20 mA to 30 mA
Warm White	3.2 V to 3.8 V	20 mA to 30 mA
Blue	3.2 V to 3.8 V	20 mA to 30 mA
Red	1.8 V to 2.2 V	20 mA to 30 mA
Green	3.2 V to 3.8 V	20 mA to 30 mA
Yellow	1.8 V to 2.2 V	20 mA to 30 mA
Orange	1.8 V to 2.2 V	20 mA to 30 mA
Pink	3.2 V to 3.8 V	20 mA to 30 mA
UV	3.2 V to 3.8 V	20 mA to 30 mA

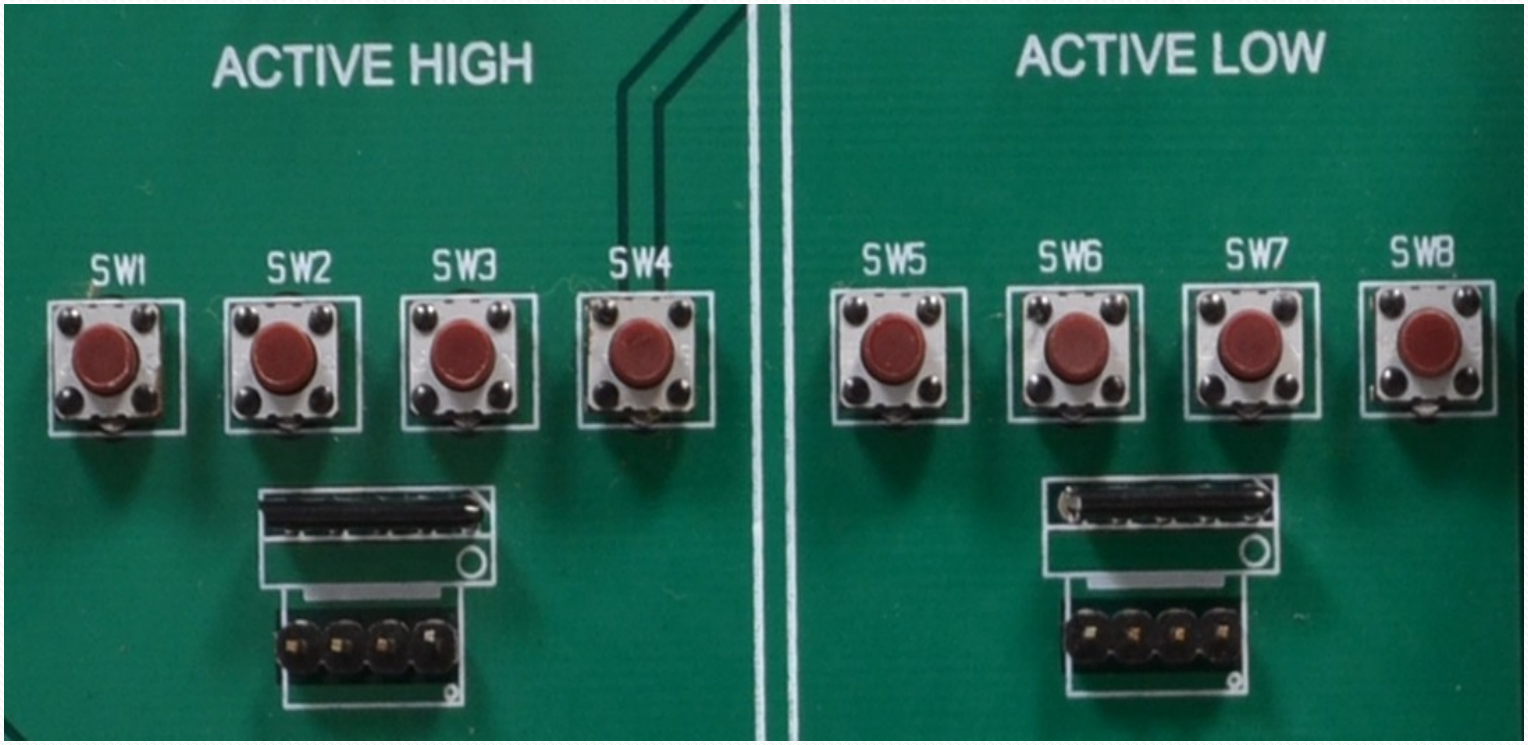
IDENTIFICATION METHODS FOR LED



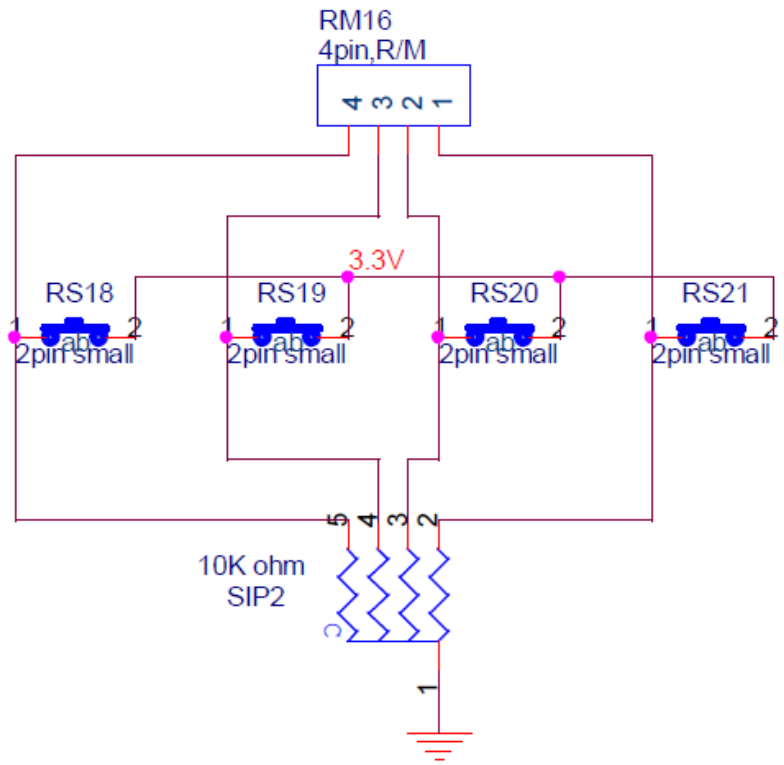
RGB LED


Flat Side
Red
Cathode
Blue
Green
Common Cathode
Lip Side
Green
Blue
Anode
Red
Common Anode
Green Anode (+)
Common
Red Anode (+)

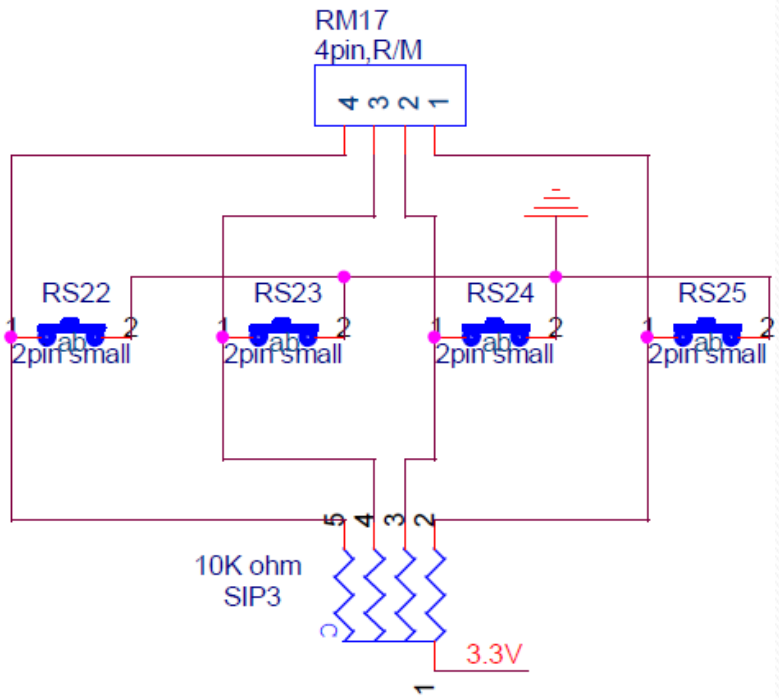
SWITCHES



SWITCHES



ACTIVE HIGH



ACTIVE LOW

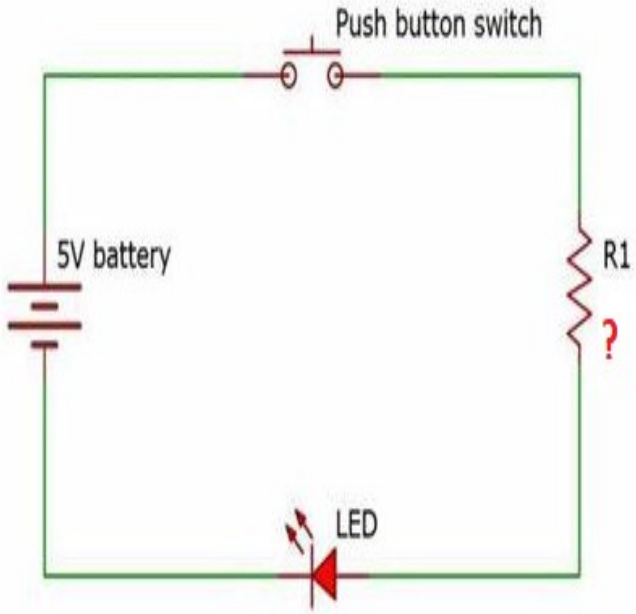
网络电阻（电阻排）、网络电容（电容排）



深圳市新祥瑞电子有限公司

注：1、黑色为RNII-6型网络电阻（电阻排）
2、黄色为CA型网络电容（电容排）
3、白色为JS双列插网络电阻

SWITCH WITH LED CIRCUIT



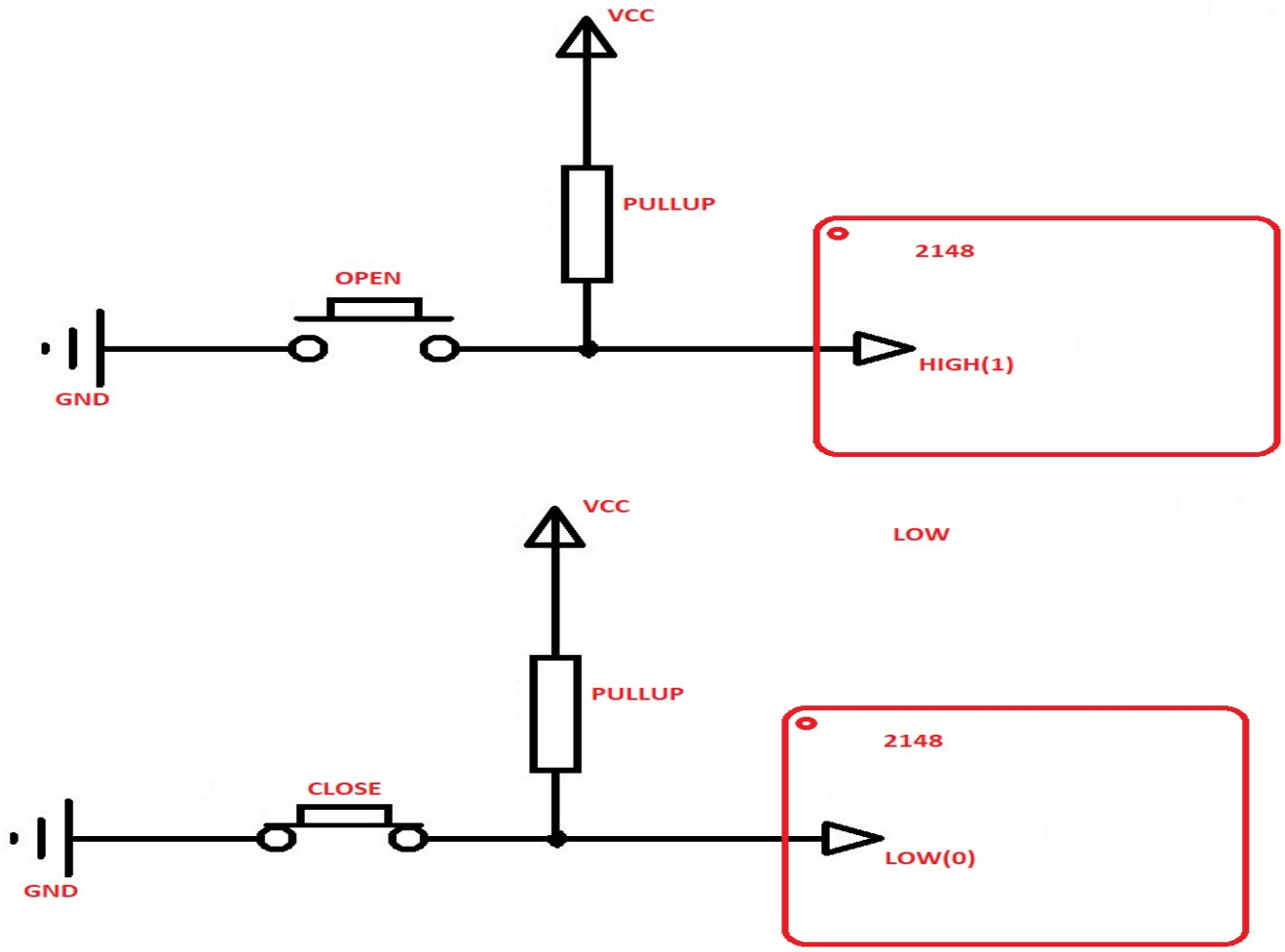


2 PIN SWITCH

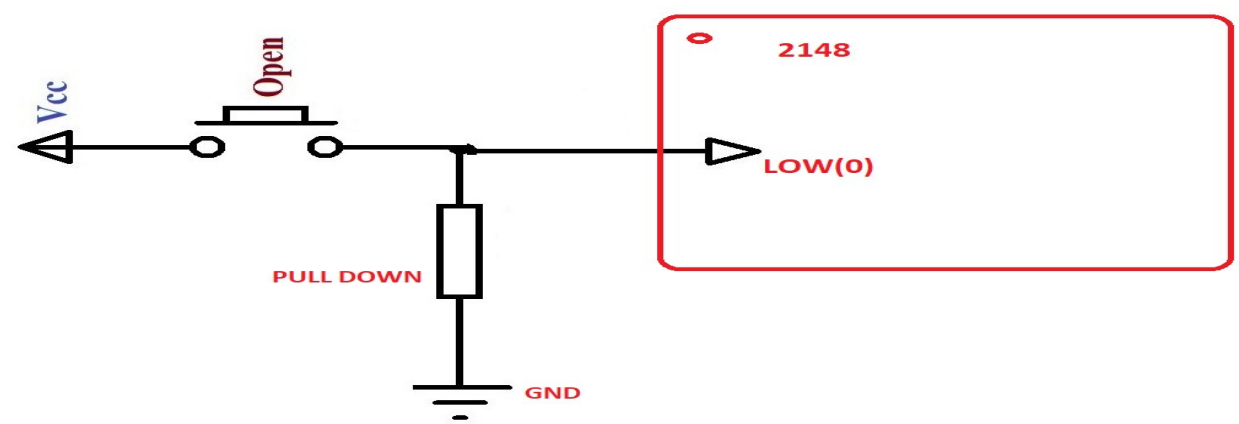


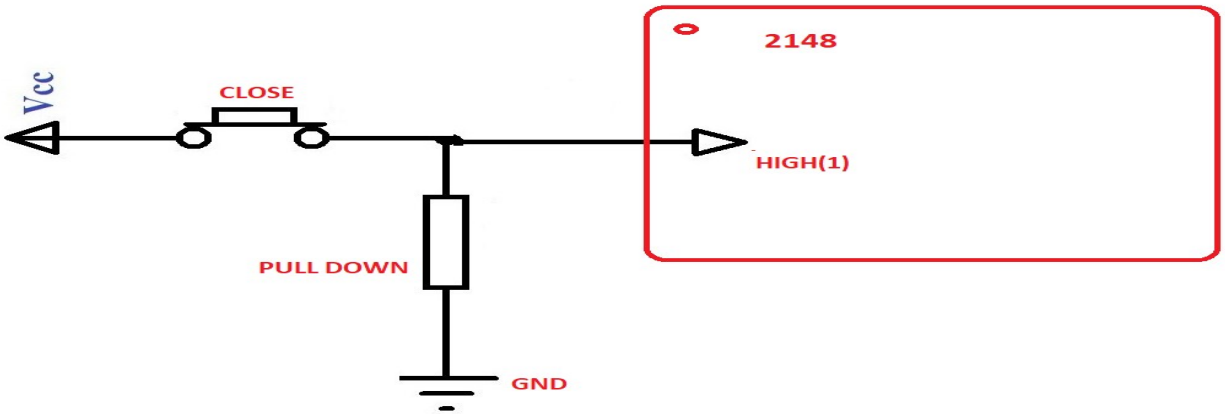
LED

ACTIVE LOW SWITCH WITH PULLUP RESISTOR

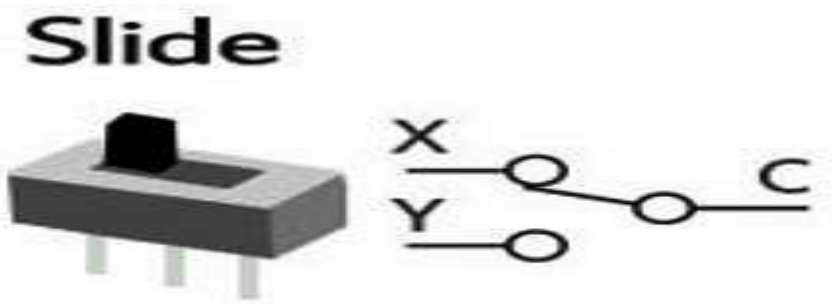


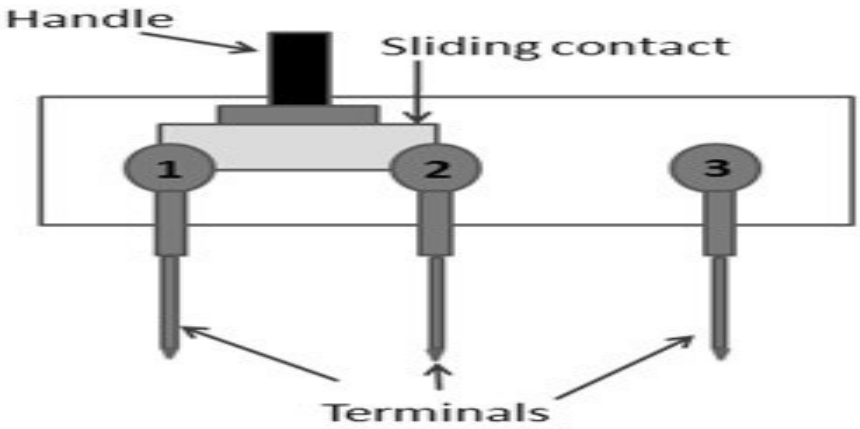
ACTIVE HIGH SWITCH WITH PULLDOWN RESISTOR



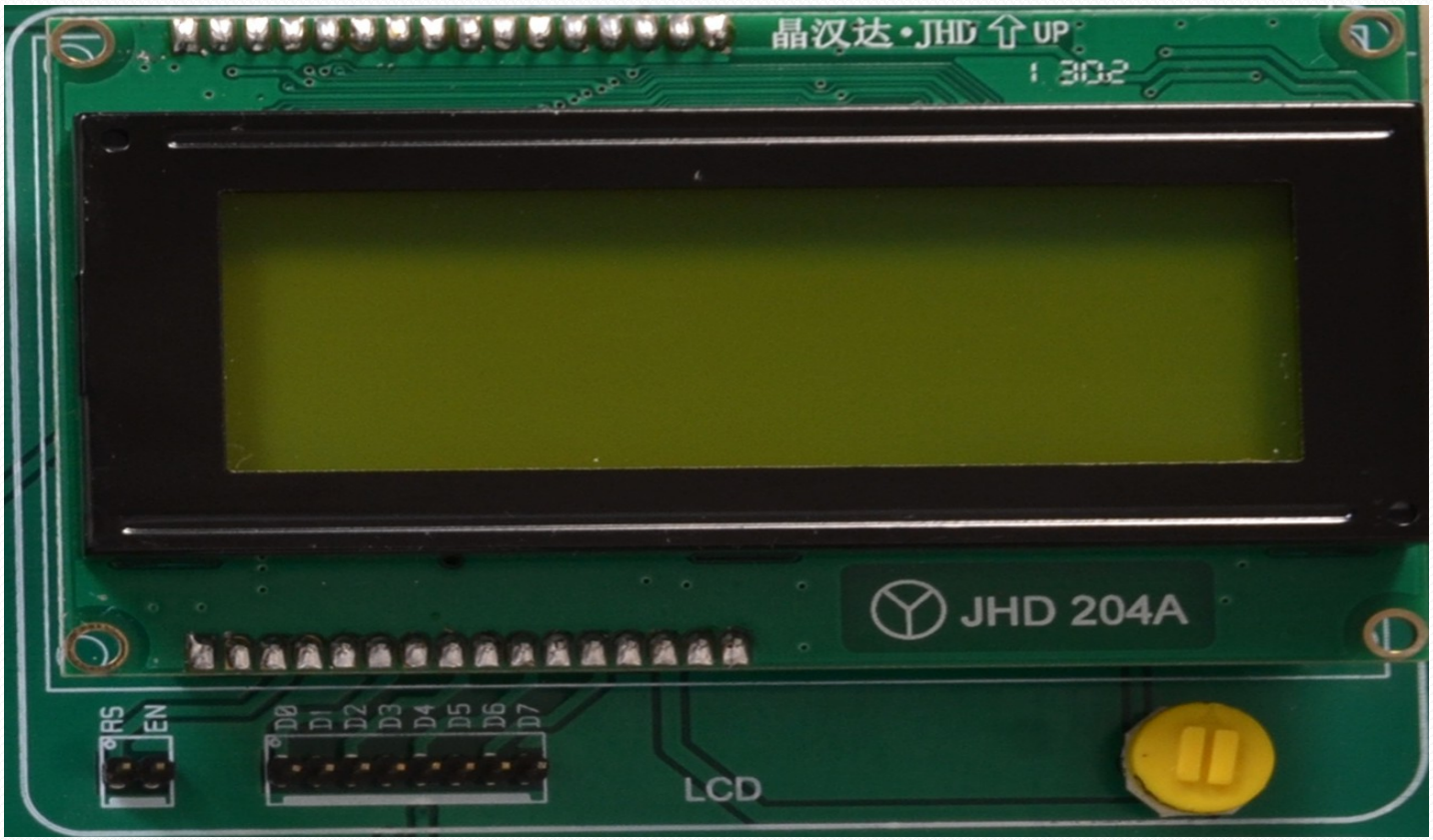


SLIDE SWITCH

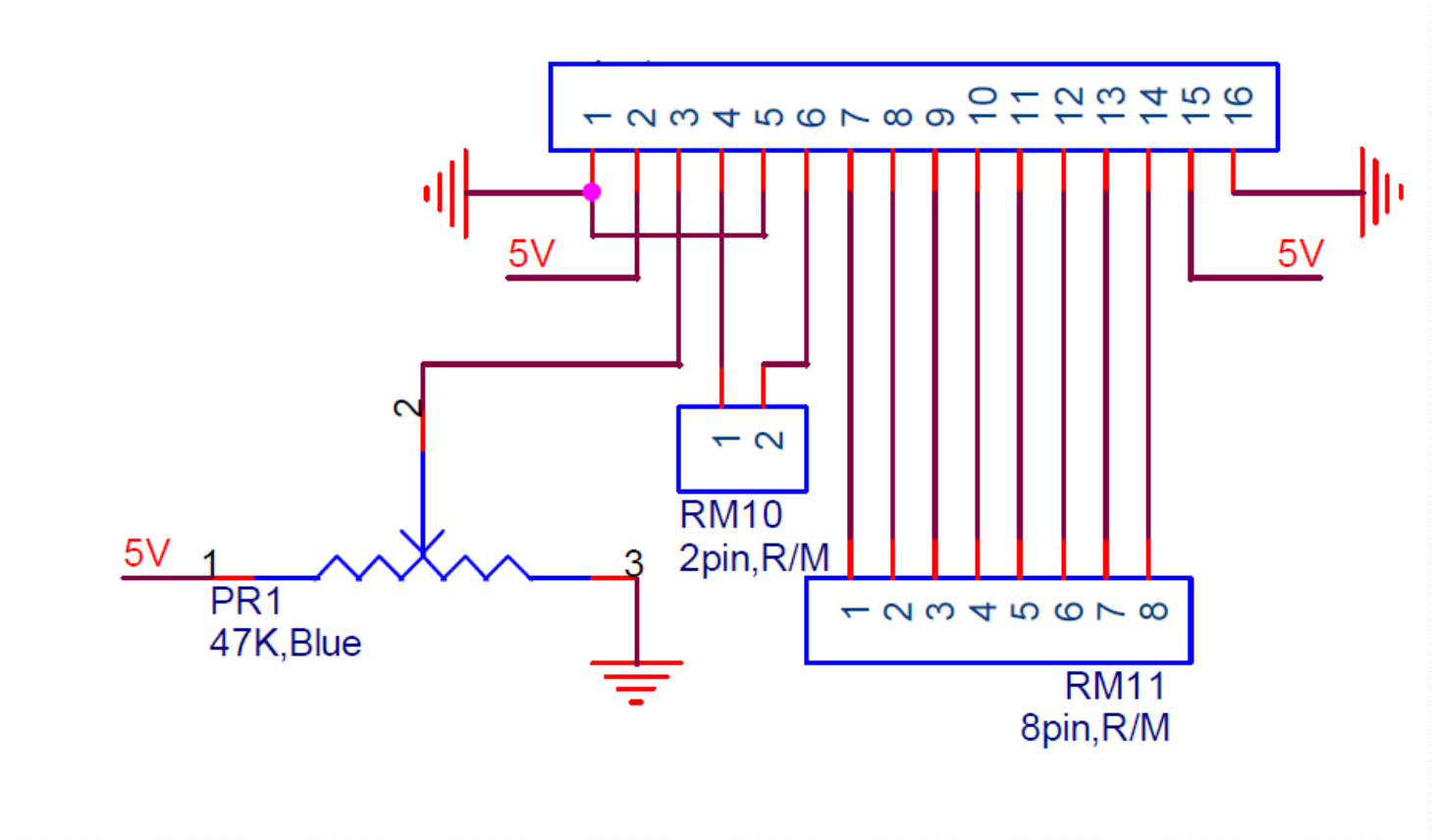




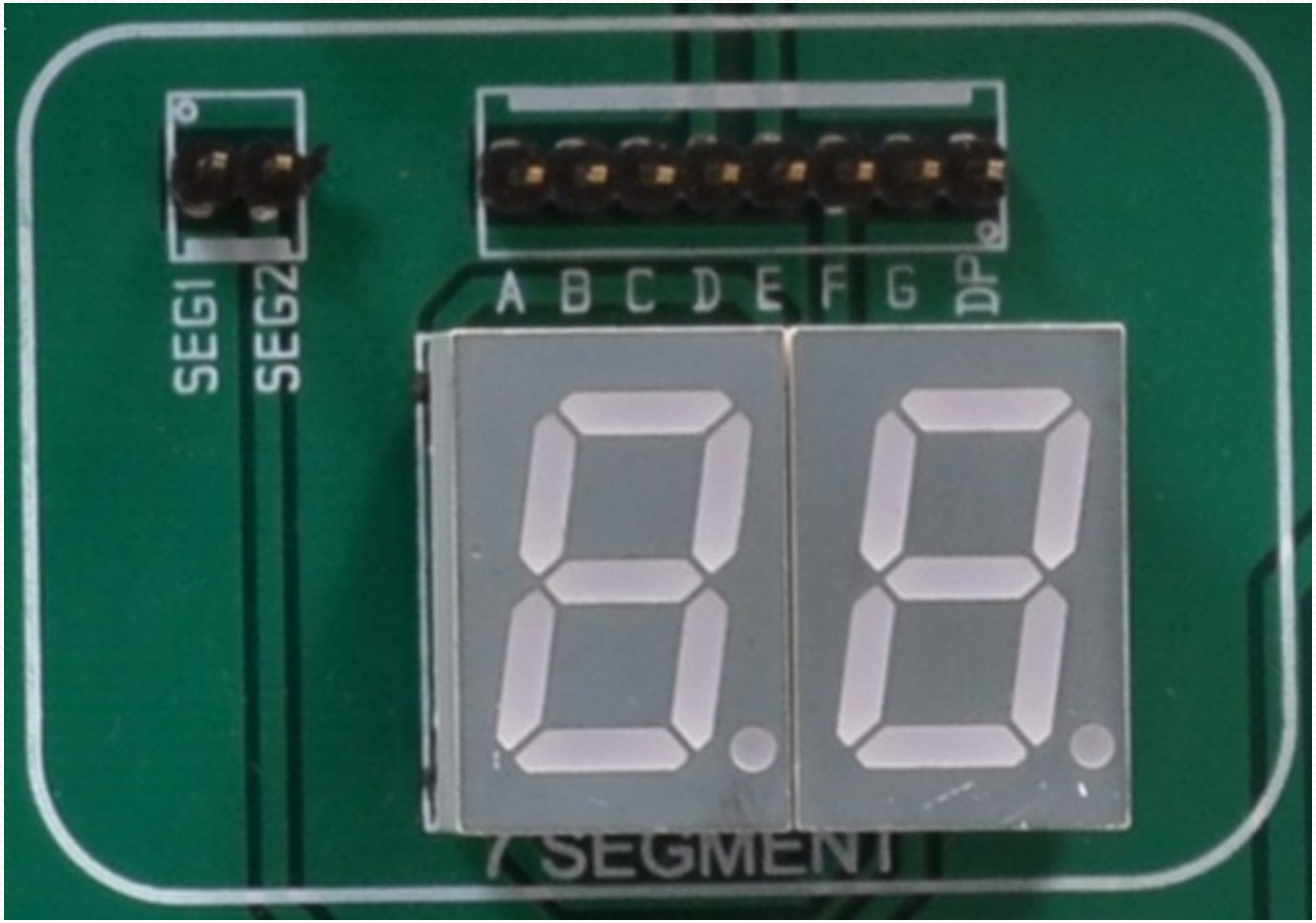
LCD



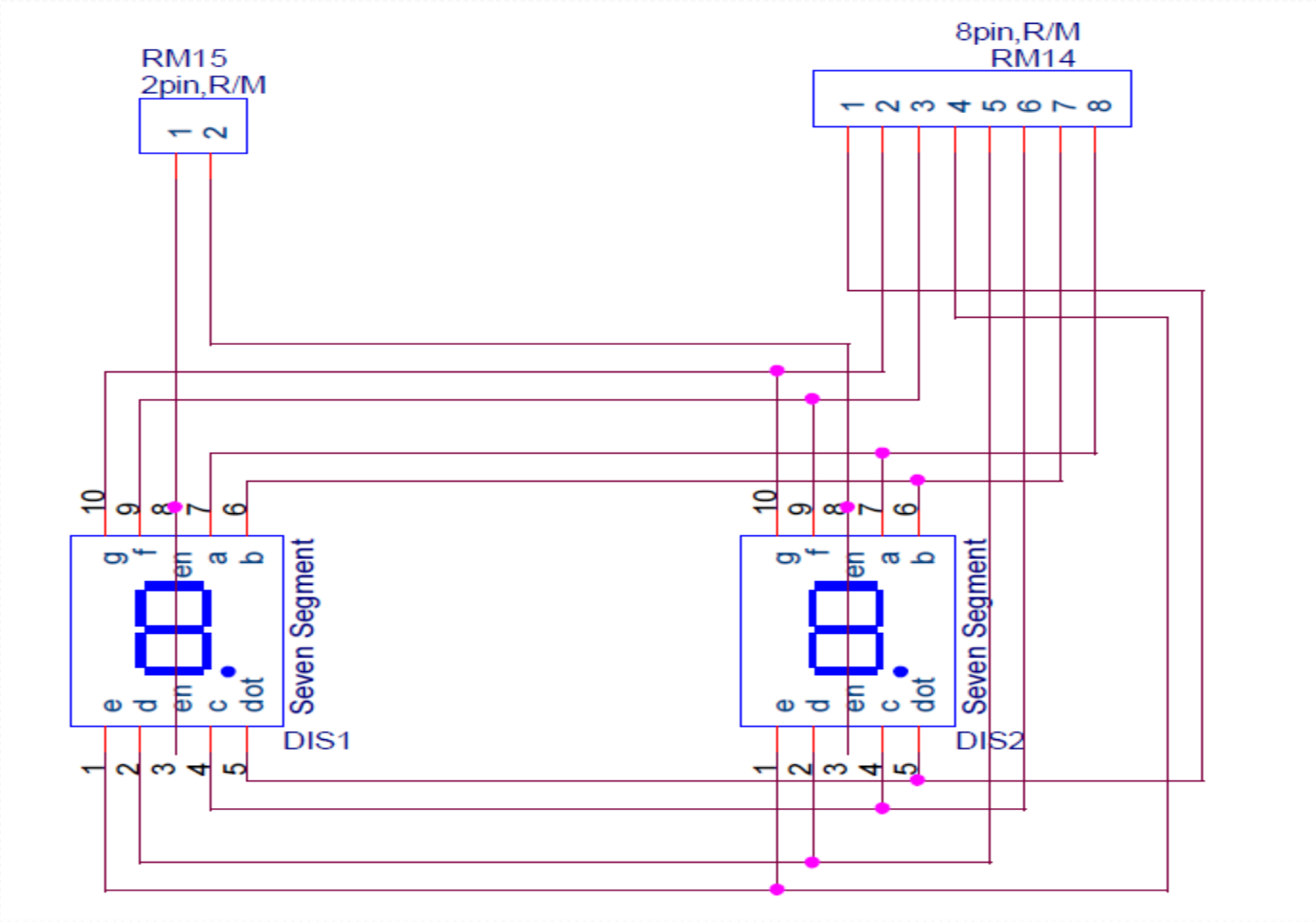
LCD


1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
5V
5V
2
1 2
RM10
2pin,R/M
5V
1
PR1
47K,Blue
3
1 2 3 4 5 6 7 8
RM11
8pin,R/M

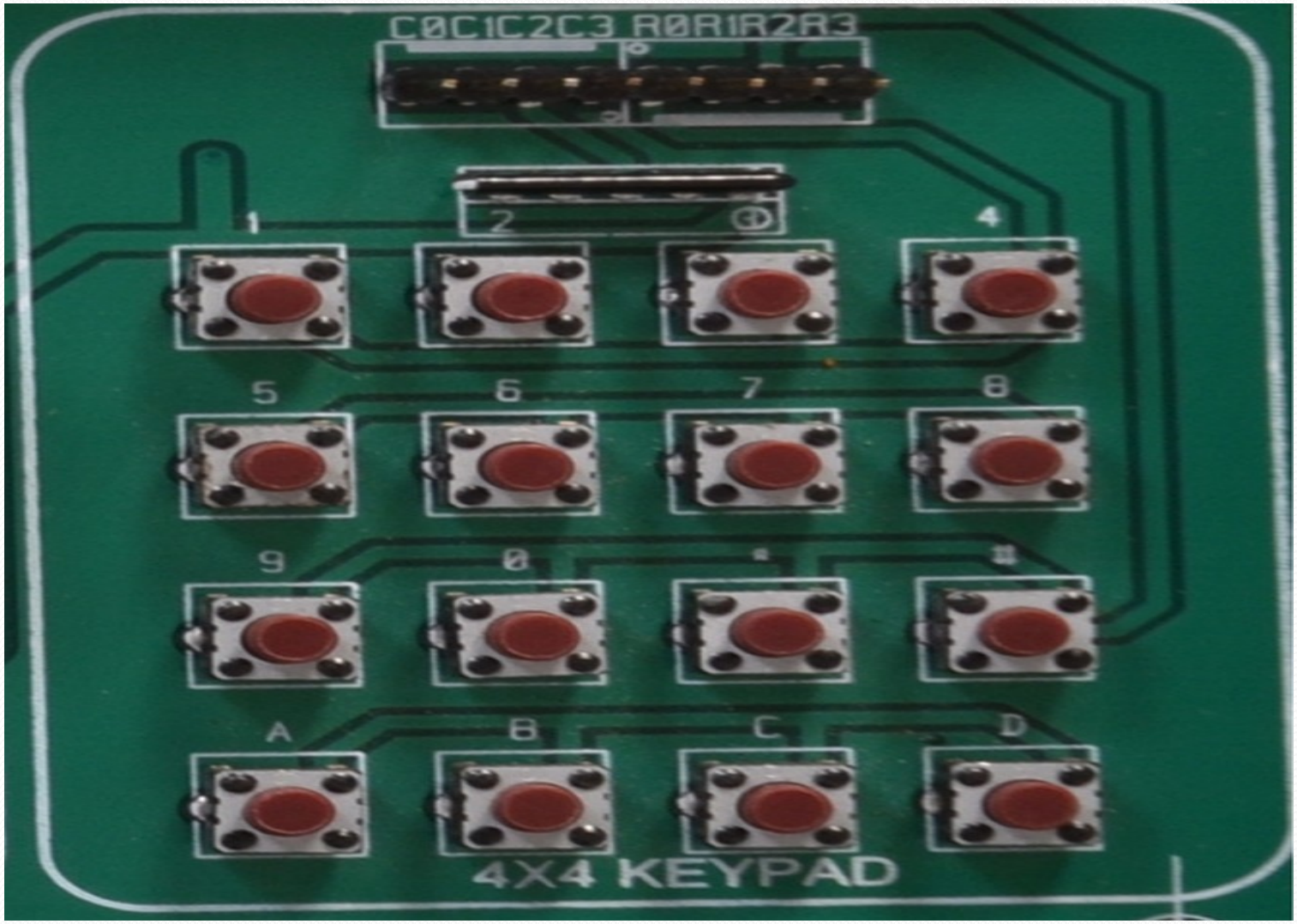
SEGMENTS



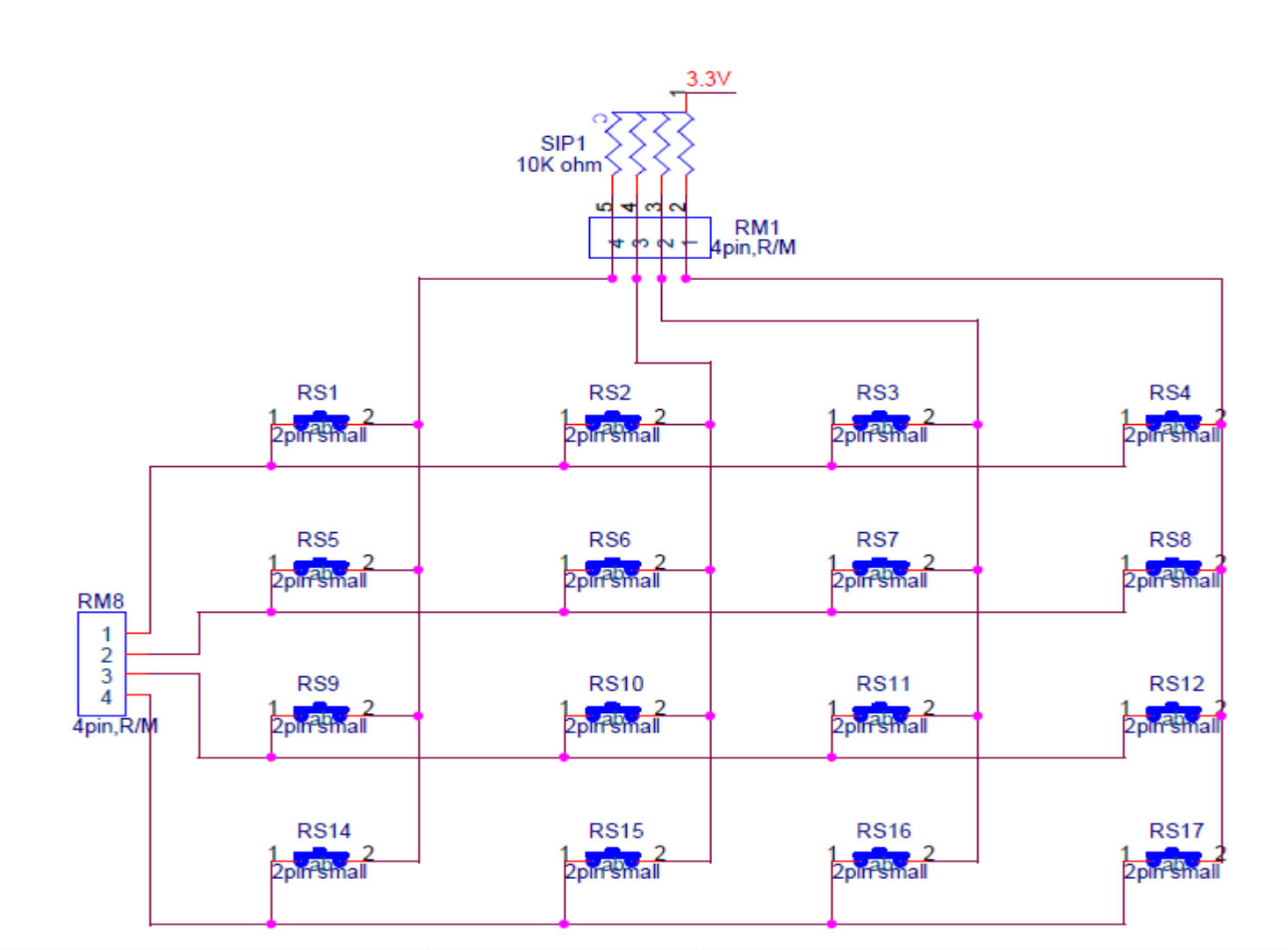
SEGMENTS


RM15
2pin,R/M
1
2
8pin,R/M
RM14
1
2
3
4
5
6
7
8
10
9
8
7
6
g
f
en
a
b
e
d
en
c
dot
Seven Segment
DIS1
1
2
3
4
5
DIS2

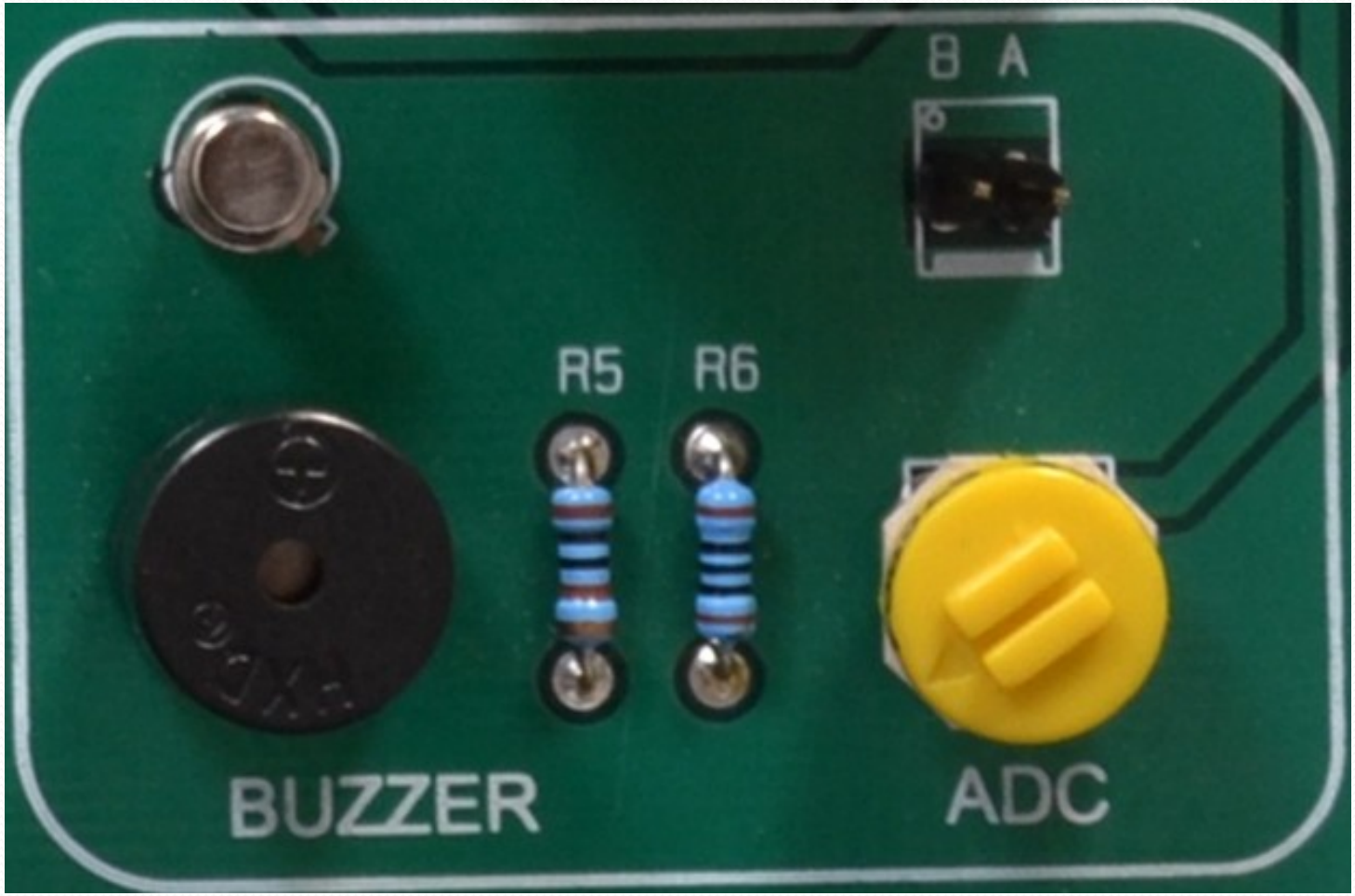
4*4 KEYPAD



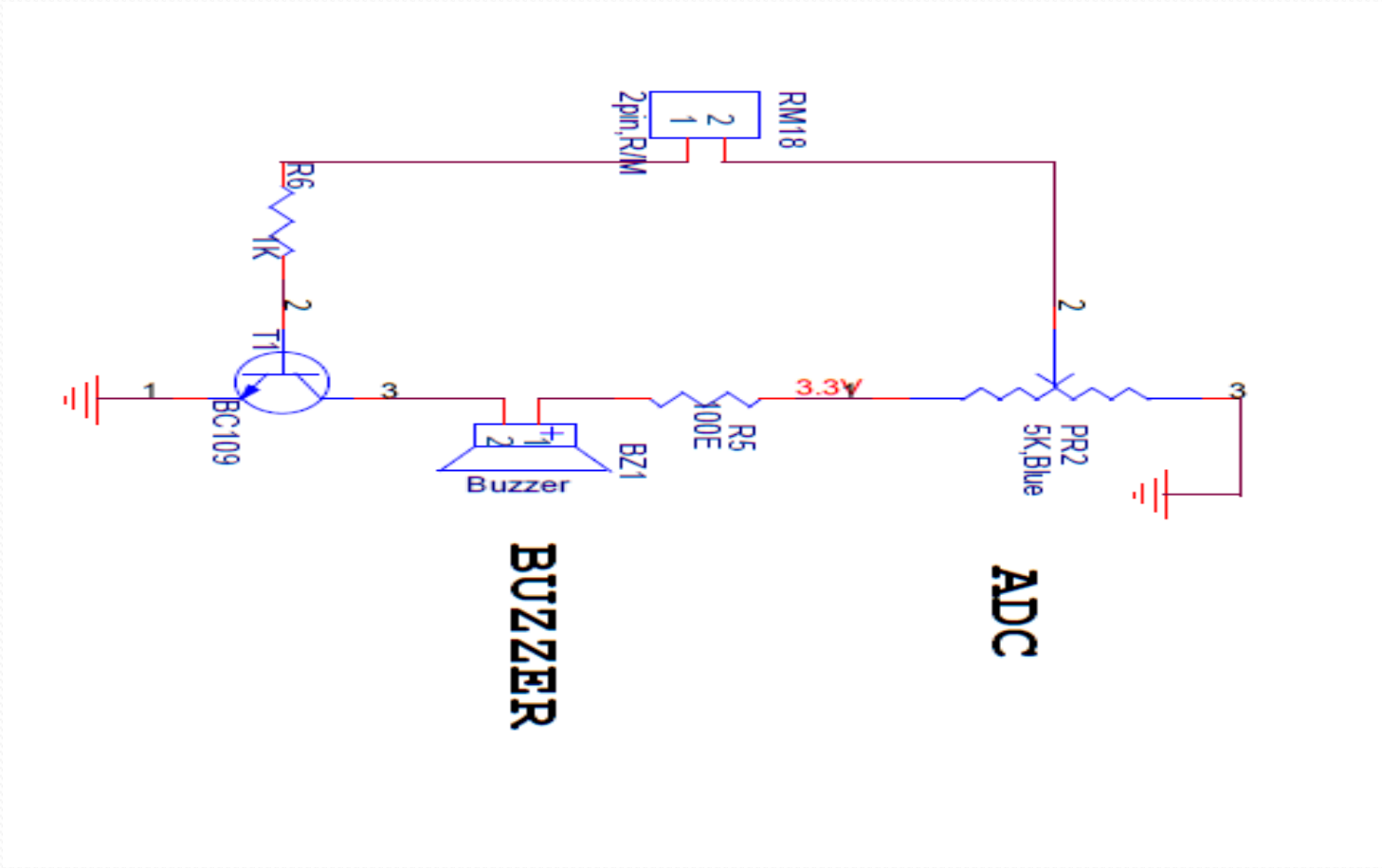
4*4 KEYPAD


3.3V
SIP1
10K ohm
RM1
4pin,R/M
RM8
4pin,R/M
RS1
RS2
RS3
RS4
RS5
RS6
RS7
RS8
RS9
RS10
RS11
RS12
RS14
RS15
RS16
RS17
2pin small

BUZZER & POT(ADC)



BUZZER **POT(ADC)**



THANK YOU